

Operations — Mini-moi

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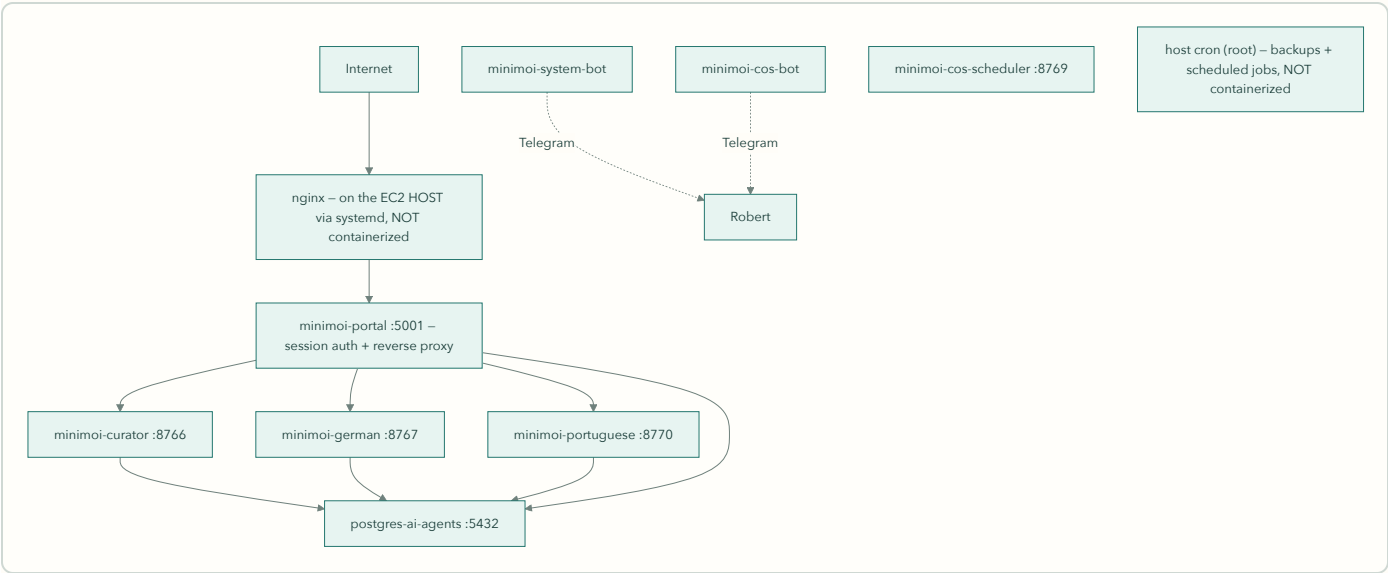
Maintained baseline – reviewed through 2026-07-21. The production topology and scheduled-job inventory were verified against the running system on 2026-07-18; the German identity and Lesen refresh updates were verified in production on 2026-07-20. The restored AI Observations path was manually verified in production on 2026-07-21; its automatic daily and Sunday runs still await verification. Later claims are marked where they rely on repository state rather than another live production inspection. Companion evidence file: `CURRENT_STATE_OPERATIONS_2026-07-18.md`. System design and the reasoning behind it: `ARCHITECTURE.md` – this document is about keeping that system running.

Topology — what is actually running

Mini-moi runs across two nodes with distinct roles (see ARCHITECTURE.md for the design rationale; this is the operational view):

Node	Role	URL	Compose file
AWS EC2 (t3.small, us-east-1, i-0d13db821169627e2)	Production – live traffic, all bots, scheduled jobs	minimoi.ai	<code>docker-compose.prod.yml</code>
Mac (local, Colima)	Dev / standby – development, private-repo sync, DNS-switchable fallback	dev.minimoi.ai	<code>docker-compose.yml</code>

Production: 8 containers + 2 host-level services



Container	Port (localhost)	Purpose
postgres-ai-agents	5432	PostgreSQL (<code>personal_agents</code>) – auth, guild, portuguese, research, pipeline, jobs schemas
minimoi-portal	5001	Session auth + reverse proxy – single entry point; forwards identity headers, domains never trust client-supplied identity
minimoi-curator	8766	Curator Flask service
minimoi-german	8767	Mein Deutsch Flask service
minimoi-portuguese	8770	Meu Português Flask service
minimoi-system-bot	–	Telegram polling bot – inbound system/German commands only; the Curator briefing is sent by <code>telegram_bot.py --send</code> inside <code>minimoi-curator</code> , via cron, on the separate outbound token
minimoi-cos-bot	–	CoS chat bot
minimoi-cos-scheduler	8769	CoS scheduled-loop agent

Two things run on the host, outside Docker – easy to forget, previously documented nowhere:

1. **nginx** – native systemd service (active since 2026-06-20), reverse proxy in front of the portal container. `systemctl status nginx` to check; config under `/etc/nginx/`.
2. **cron (root)** – all backup jobs and scheduled pipelines. `sudo crontab -l` to see the real schedule. A container-only mental model of production misses both of these.

All container ports bind to `127.0.0.1` – nothing is directly exposed except through nginx.

Deploy & Rollback

Deploy

Push to `main` triggers CI/CD (GitHub Actions): test → build images → push to ECR → deploy to EC2 via SSM. **~4-5 minutes to live.**

Two operational facts worth stating plainly:

- **A plain `git push` to `main` rebuilds and redeploys all 7 built services automatically – including on docs-only commits.** There's no path filtering in the workflow, and a fresh CI build always produces a new image digest even for byte-identical source, so `pull && up -d` recreates all 7. **postgres-ai-agents is the one exception** – stock image, not in the build matrix, never touched by the pipeline. Live proof: after a docs-only push, `docker inspect` showed all 7 app containers sharing an identical start timestamp to the second, while postgres kept its prior-day timestamp. All 7 restart *together*, not staggered – each comes back in seconds, so in practice it reads as near-zero perceived downtime rather than a visible outage. But it is all 7, not just the one you happened to be using. Updating a *single* container is possible, but only via the

manual path under Rollback below – it is not something a push triggers. One footnote: the workflow's `workflow_dispatch` trigger defines an `image_tag` input that the deploy job never actually reads – it always pulls `:latest`. That input is vestigial; don't mistake it for a working single-tag deploy capability.

- **There is no staging gate in the deploy pipeline.** A push to `main` goes straight to production. `dev.minimoi.ai` routes via Cloudflare Tunnel to the Mac dev environment (verified in the tunnel config) – a genuinely separate origin, useful for development, but not a gate the pipeline passes through on the way to prod. A real AWS staging environment is committed work (see ROADMAP). Until then: the standing rule is that nothing is pushed without explicit approval of the reviewed diff.

Recovery options

Access channels (always available, independent of any rollback action): SSH (security-group allowlisted) and AWS SSM Session Manager / EC2 Instance Connect – the fallback that works even when SSH rules are wrong. Both verified working.

Rollback actions, typical order:

1. **git revert + push** – rides the same CI/CD pipeline; previous state live in ~4-5 minutes. The default for any bad deploy.
2. **Manual container recreate on the host** – can be faster than a revert when only one service is affected:

```
cd /opt/minimoi
# Verify which Compose form this host actually has before an incident, not during:
#   which docker-compose | docker compose version
sudo docker-compose -f docker-compose.prod.yml up -d --force-recreate <service-name>
# Use the Compose SERVICE name (e.g. curator), not the container name.
```

Pulling a specific prior image from ECR is possible but Compose is pinned to `:latest` – an exact, tested command for that path is an open runbook item, not yet written.

Manual health check (EC2)

```
sudo docker ps --format "table {{.Names}}\t{{.Status}}"
df -h / # 30GB volume; alert threshold 80%
free -m
curl -s http://localhost:8766/health # curator
curl -s http://localhost:8767/health # german
curl -s http://localhost:8770/health # portuguese
curl -s http://localhost:5001/health # portal
systemctl status nginx # host nginx – not in docker ps
sudo docker logs minimoi-portal --tail 30 # when something looks wrong
```

Automated version: CoS's Loop H runs every 30 minutes – checks the seven other expected containers (it cannot meaningfully check its own liveness from inside itself), disk < 80%, memory < 85%, and four /health endpoints – alerting via the **CoS bot token** (not the system bot). It does **not** check host nginx; nginx is currently

unmonitored beyond systemd itself. CoS detects and escalates – **Robert decides the action.**

Scheduled Jobs

Cron, split across two users, is the entire scheduling story on this host – verified live 2026-07-18 (`crontab -l` as both users; `systemctl list-timers` confirmed no app-level systemd timers exist, only stock OS ones). Checking one crontab and assuming it's the whole picture misses the pipeline.

User	Time (UTC)	Job	What it does
root	02:00 daily	<code>backup_local.sh</code>	Tier 1 – full local backup to <code>/opt/minimoi/backups/YYYY-MM-DD/</code>
root	03:00 daily	<code>backup_s3.sh</code>	Tier 2 – S3 sync
root	04:00 Sunday	<code>backup_dropbox.sh</code>	Tier 3 – weekly Dropbox sync – BROKEN: rclone not installed; fails silently every run
ec2-user	hourly	<code>run_curator_cron_ec2.sh</code>	Curator pipeline – fires at most once/day, gated below
ec2-user	15 minutes past each hour	<code>run_intelligence_cron_ec2.sh</code>	AI Observations – waits for the daily briefing, runs at most once/day, and adds the weekly synthesis on Sunday; installed and manually verified 2026-07-21, automatic runs awaiting verification

German's Lesen refresh command is deployed and was exercised successfully on 2026-07-20, adding 35 current articles while reporting two failed sources without writing a false success marker. **It is not yet installed as a production cron job.** Until that final scheduling step is completed, current reading material still requires an explicitly approved refresh.

The pipeline wrapper runs hourly but only actually fires once a day, gated in order: **role guard** (exits silently unless `MINIMOI_ROLE=production` inside the container – safe on a standby node), **time gate** (12:00-20:00 UTC only), **idempotency check** (reads `briefing_date` from `curator_latest.json`; skips if today already ran). When it fires: pulls new X bookmarks (non-fatal on failure), runs the scoring pipeline, sends the Telegram briefing, stamps the date. One known wart, documented in ARCHITECTURE.md and tracked separately: the `--model=grok-4.3` value the script passes isn't valid and silently falls through to the hardcoded default – the model that runs is right by coincidence, not by configuration.

In-container scheduling (APScheduler) — the second scheduling layer

Host cron is not the whole story. `minimoi-cos-scheduler` runs its own in-process scheduler with **seven jobs** (confirmed in `domains/cos/chief_of_staff.py`):

Job	Schedule
Loop A – career focus scout	06:00 and 18:00 daily
Loop B – German watch	Sunday 09:00
Loop C – Curator watch	Sunday 10:00
Loop D – novelty watch	1st and 15th, 08:00
Loop F – build-discipline check	daily 07:30
Loop G – guest-access staleness nudge	hourly
Loop H – EC2 health check	every 30 min

The scheduler's timezone has not been explicitly confirmed – verify before relying on these clock times. Checking only `crontab -l` misses this entire layer; the full scheduling inventory is host cron (both users) **plus** this container scheduler.

Data Persistence & Backups

Persistence

All three domain containers mount their data directories to the host – confirmed in `docker-compose.prod.yml`. Container restarts and redeploys do not touch domain data. This closed the ephemeral-storage defect that cost the June 22–July 13 curator briefings (lost before the mounts existed – a mounts problem, historical and unrecoverable, not an ongoing risk).

Also host-mounted: `guests.json`, `build_queue.json`, `cos_memory.md`, `cos_context.json`, `docs/design/`, `docs/specs/`, and `/opt/minimoi/agent_logs/`.

Backups — Tiers 1–2 live, Tier 3 broken

- **Tier 1 (local, 02:00 daily):** confirmed working – dated folders in `/opt/minimoi/backups/` through today. What it captures: full `data/` rsync, auth files, local Postgres dump, `agent_logs/` sync, retention prune. Includes a manual pre-password-rotation Postgres dump (2026-07-16).
- **Tier 2 (S3, 03:00 daily):** confirmed working – real objects in `s3://minimoi-backups/`, logs show clean completion on consecutive recent days.
- **Tier 3 (Dropbox, 04:00 Sunday): broken, never ran successfully.** `rclone` was never installed on EC2 – the job's single log entry (its first scheduled Sunday) is `rclone: command not found`, and it has failed silently every week since. It remains scheduled and will keep failing until fixed. Tier 2 has an error trap; Tier 3 has no working failure alert.

Status corrections, both on the record: defect #136's framing of Tier 1 as "never built" was wrong – `scripts/backup_local.sh` exists and its cron runs; that half of #136 should be formally closed with a correcting comment. And the first version of *this* document claimed all three tiers verified – that verification checked cron entries and folder structure but not log output, which is exactly how a silently failing Tier 3 passed it. Verified means logs and outcomes, not schedules and folders.

Known gaps that remain true: Tier 3 needs `rc1one` installed and a first successful run; Tier 3 failure does not alert; and a backup that exists is not a backup that restores – no restore test has been run. All tracked below; the restore test is planned as the acceptance test for the staging environment.

Monitoring

Layer	Status	Detail
Sentry	Live	Wired into curator, german, portuguese servers and the portal (shipped 2026-06-23)
CoS health loop	Live	Every 30 min on EC2: containers, disk, memory, /health endpoints → Telegram alert
CloudWatch	Live (basic)	EBS disk cross-check used by the health loop
Prometheus / Grafana	Proposed, never built	Exists only in spec documents. Do not treat any historical monitoring-stack spec as describing production.

Known monitoring gaps, tracked: silent Postgres failure during login `auth_id` lookup (#84 – `except Exception: pass` with no logging), and no break-glass admin account (#87).

Credentials & Third Parties

Production credentials live in AWS SSM Parameter Store under `/minimoi/production/*` – never in files, never in git. (Mac/dev uses macOS Keychain via `keyring`; that story is dev-only now.) The live parameter set is the authoritative third-party inventory:

Category	Parameters
LLM APIs	<code>grok_api_key</code> (xAI), <code>anthropic_api_key</code> , <code>openai_api_key</code>
Search / retrieval	<code>brave_api_key</code> , <code>tavily_api_key</code>
Translation	<code>deepl_api_key</code>
Messaging (Telegram)	<code>telegram_bot_token</code> , <code>telegram_cos_bot_token</code> , <code>telegram_system_bot_token</code> , <code>telegram_polling_bot_token</code> , <code>telegram_agent_bot_token</code>
Email	<code>zoho_smtp_password</code> (live – portal email), <code>gmail_app_password</code> (orphaned – zero code references; resolve: delete or wire up)
App / DB	<code>flask_secret_key</code> , <code>postgres_password</code> , <code>minimoi_password</code> , <code>minimoi_agent_password</code> , <code>robert_sql_password</code>

Telegram token → bot mapping (traced to consuming code, 2026-07-18):

SSM parameter	Consumed by	Runs where
telegram_bot_token	telegram_bot.py – outbound briefing sender	minimoi-curator (invoked by hourly cron)
telegram_polling_bot_token	telegram_bot.py – inline-button/callback handling	minimoi-curator
telegram_system_bot_token	telegram_system_bot.py – own polling loop	minimoi-system-bot
telegram_cos_bot_token	telegram_cos_bot.py	minimoi-cos-bot
telegram_agent_bot_token	utils/telegram.py – OpenClaw gateway	Mac only – per its own code comment

One placement note: `telegram_agent_bot_token` is scoped in code to the Mac-only OpenClaw gateway, yet it sits in the `/minimoi/production/` SSM namespace alongside the four that actually run in production. Not broken – misfiled. Cleanup candidate. One residual thread: whether `telegram_polling_bot_token` is actively exercised in production today (vs. a webhook-testing leftover) was not verified in this pass.

One real exception to the SSM pattern: Curator's production xAI scorer reads its key directly from `~/.openclaw/agents/main/agent/auth-profiles.json` – an OpenClaw-managed file at a fixed path – rather than the shared SSM helper every other service uses (German, Portuguese, CoS all resolve `xai_api_key` via the helper). This is fragile and undocumented anywhere else: the Curator container depends on that file existing. Related naming wrinkle to reconcile: the SSM inventory lists `grok_api_key` while the shared helper resolves `xai_api_key` – verify how the live container actually obtains the key, then document one authoritative name.

DB roles are separated (`robert_sql` , `minimoi_agent`) and rotated off the old weak password – confirmed as distinct SSM parameters.

Docs Sync

Committed docs reach the production doc viewer via `scripts/sync_docs.sh` : git commit → GitHub raw URL → SSM pull to the host (docs paths are host-mounted into the portal container). **Exact scope, verified in the script:** `docs/design/`*, `docs/specs/`*, and `data/guild/build_queue.json` – and nothing else. Root documents (`ARCHITECTURE.md` , `OPERATIONS.md` , `ROADMAP.md`) are **not** synced: they are GitHub-only unless a copy is placed under `docs/design/` or the sync script is extended. **A git push alone does not make a new spec visible in prod** – the sync step is what lands it.

Dev / Standby Environment (Mac)

The Mac runs substantially more than a mirror of the prod compose stack – this section is the fully verified picture (live `launchctl` / `docker` / `crontab` / `lsuf` inspection, 2026-07-18), and most of it was previously documented nowhere.

Application layer actually running on the Mac

Service	Port	Runs as	Notes
Guild Operations agent	8768	launchd <code>com.user.operations</code>	Guild's agents appear only here – absent from the prod container list entirely
Guild Dev agent	8771	launchd <code>com.user.devagent</code>	Same
CoS (bare)	8769	launchd <code>com.user.cos</code>	Runs simultaneously with the containerized dev CoS below – dual-instance
CoS (containerized dev)	18769	Docker <code>minimoi-cos-dev</code>	
Portuguese	8770	launchd <code>com.user.portuguese</code>	Plus scheduled <code>com.user.portuguese-leitura</code>
German	8767	launchd <code>com.vanstedum.german-html-server</code>	Plus hourly time-gated <code>com.vanstedum.lesen-refresh</code>
System bot (Mac-local standby)	–	launchd <code>com.vanstedum.system-bot</code>	Distinct from the EC2 system-bot container – code implements a deliberate standby/production token switch so the two never collide on the same Telegram token
cloudflared, Colima	–	launchd	Tunnel + Docker runtime
Usage report	cron 08:00 + 10:00	<code>scripts/track_usage_wrapper.sh</code>	No matching crontab entry was present on the Mac when checked 2026-07-26; restore deliberately if this report is still wanted

Dev-only by design, never migrating: Ollama (local models, `gemma3:1b`), the OpenClaw gateway, and the nightly private-repo sync (`scripts/sync_private_repo.sh` , 02:00 local). Keychain-based credentials are the dev flow; production uses SSM.

Legacy automation still loaded: five launchd jobs from the pre-AWS era remain loaded but idle (`com.vanstedum.curator` , `curator-intelligence` , `curator-priority-feed` , `portal-boot-restart` , `minimoi-portal`). Not flagged for removal here – flagged so nobody is surprised that the Mac has more loaded automation than the active-service list suggests.

Open item – port 8766 ambiguity: a native launchd process (`com.user.curator-server` , running `curator_server.py` via the project venv) holds host port 8766 per `lsof` , while the local `minimoi-curator` container claims to publish the same port and `docker port` reports the mapping as active. The OS-level listener answering `curl localhost:8766/health` is the *native* process. Which one Docker-bound traffic actually reaches was not resolved – tracked in Known Gaps, stated here as an ambiguity rather than a conclusion.

After a Mac reboot: `colima status` → `docker ps` → start what's missing. Dev being down never affects production.

Known Gaps & Open Items

Item	Tracked	Status
Curator cron wrapper masks failed scoring runs – <code>STATUS=\$?</code> captures Telegram's exit code, not curator's; a failed run + successful stale send stamps <code>briefing_date</code> as success	–	High – idempotency guarantee weaker than documented; found by Codex review 2026-07-18
Tier 3 Dropbox backup broken – rclone never installed; fails silently weekly	–	High – fix before the restore test; found by review 2026-07-18
Curator xAI key read from an OpenClaw file path, outside the SSM pattern; SSM naming (<code>grok_api_key</code> vs <code>xai_api_key</code>) unreconciled	–	Open – verify live retrieval path, then standardize
Host nginx unmonitored (health loop doesn't check it)	–	Open
AI Observations production scheduling	OpenClaw issue	Schedule installed and manual daily run verified 2026-07-21; automatic daily and Sunday weekly runs still require production confirmation
German Lesen refresh command is deployed and production-tested but has no production schedule	#96 follow-up	Open – install and verify cron separately
APScheduler timezone unconfirmed for the 7 in-container jobs	–	Open – verify before relying on clock times
Tier 3 backup failure does not alert	–	Open – same silent-failure family as #84
No break-glass admin account	#87	Open
Silent Postgres failure on login lookup (no logging)	#84	Open – next after current work per standing decision
No isolated staging environment (dev = prod origin)	–	Acknowledged; future AWS staging
Backup restore test never run	–	Open – instrumented follow-up
<code>telegram_agent_bot_token</code> in production SSM namespace but Mac-only in code	–	Misfiled, not broken – naming/placement cleanup
<code>telegram_polling_bot_token</code> – confirm actively used in prod vs. leftover	–	Open – one residual from token mapping
Mac port 8766: native <code>curator_server.py</code> vs. Docker container both claim it	–	Open – resolve which one traffic actually reaches
Vestigial <code>workflow_dispatch</code> <code>image_tag</code> input in <code>deploy.yml</code>	–	Remove or document – implies a capability that doesn't exist
Orphaned <code>gmail_app_password</code> SSM parameter	–	Delete or wire up
#136 formally close with correcting comment	#136	Robert/OpenClaw action

Production topology verified 2026-07-18; German identity and Lesen refresh state verified 2026-07-20. When this document and the running system disagree, the system is right – fix the document, and say so in the commit.